

Exercise Sheet 4

Naive Bayes and the Generative Route

Probabilistic Machine Learning

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4.A The Generative Route

Exercise 4.A.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint $p(\mathbf{x} \mid y)$ per class?
2. How many does naive Bayes need?
3. For p yes/no attributes and K classes, give both counts as formulas.
4. From which p on does the full joint need more than a million numbers *per class*?

Exercise 4.A.2: Fitting Is Counting

An attribute takes k values. Within one class we observed it n times, value v exactly n_v times ($\sum_v n_v = n$). Model it as categorical with parameters $\theta_1, \dots, \theta_k \geq 0, \sum_v \theta_v = 1$.

1. Write down the log-likelihood $\ell(\theta)$ of the n i.i.d. observations.
2. For $k = 2$, substitute $\theta_2 = 1 - \theta_1$ and show that the MLE is $\hat{\theta}_1 = n_1/n$ — the h/n of Lecture 03.
3. For general k , maximise ℓ under the constraint $\sum_v \theta_v = 1$ (Lagrange multiplier) and show $\hat{\theta}_v = n_v/n$.
4. Why can naive Bayes fit every attribute's table separately, while a full joint $p(x \mid y)$ could not be fitted attribute by attribute?

4.B Naive Bayes by Hand

Exercise 4.B.1: Twelve Transactions

Flags: new device D , foreign IP F , night N , high amount H (1 = yes).

#	D	F	N	H	fraud?	#	D	F	N	H	fraud?
1	1	1	1	1	yes	7	1	0	0	0	no
2	1	1	0	1	yes	8	0	0	0	0	no
3	1	0	1	0	yes	9	0	1	0	0	no
4	0	1	1	1	yes	10	0	0	0	0	no
5	0	0	0	0	no	11	1	0	1	0	no
6	0	0	1	0	no	12	0	0	0	0	no

1. Estimate $P(\text{fraud})$ and $P(\text{flag} = 1 \mid \text{class})$ for all four flags by counting.
2. Compute $P(\text{fraud} \mid x)$ for $x_1 = (D, F, N, H) = (1, 0, 0, 0)$.
3. Do the same for $x_2 = (0, 0, 0, 1)$. What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

Exercise 4.B.2: Smoothing Is a Prior

A yes/no flag was seen h times among n frauds.

1. With a $\text{Beta}(1, 1)$ prior on $\theta = P(\text{flag} \mid \text{fraud})$, what is the posterior (Lecture 02)? Show that its **mean** is Laplace's $(h + 1)/(n + 2)$.
2. What is the posterior's **peak** for $h = 0$? What does that say about "keep the peak"?
3. "Add- α " smoothing uses $(h + \alpha)/(n + 2\alpha)$. Which prior belongs to it? Compute the estimate for $h = 0, n = 100$ and $\alpha = 0.5, 1, 10$.
4. A merchant category with $k = 10$ values; one category never occurred in 100 frauds. What does Laplace smoothing give, and which prior is behind it?

4.C Generative vs Discriminative

Exercise 4.C.1: The Same Line

Naive Bayes with yes/no flags, $\theta_{jy} = P(x_j = 1 \mid y)$, prior $\pi = P(y = 1)$. Lecture numbers: $\pi = 1/1000$; new device 0.6 vs 0.1; foreign IP 0.7 vs 0.05; night 0.5 vs 0.2 (fraud vs honest).

1. Show that $\log \frac{P(y=1|x)}{P(y=0|x)} = w_0 + \sum_j w_j x_j$ and give w_j and w_0 .
2. Compute the three weights and w_0 , and check that a transaction with all three flags gets $P(\text{fraud} \mid x) \approx 17.4\%$.
3. The foreign-IP flag is **missing**. Compare what naive Bayes should do with plugging $x_{\text{IP}} = 0$ into the linear form.
4. Why can logistic regression, which has the same linear form, not do what naive Bayes does in 3?

Exercise 4.C.2: Counting Twice, in Code

Complete the three marked lines (numpy only), then run the script.

```
import numpy as np
rng = np.random.default_rng(42)
n = 5000
y = (rng.random(n) < 0.3).astype(int)          # 1 = fraud
flag = lambda pf, ph: (rng.random(n) < np.where(y == 1, pf, ph)).astype(int)
device = flag(0.6, 0.1)
copy = np.where(rng.random(n) < 0.95, device, 1 - device)  # a near-copy
night = flag(0.5, 0.2)
X = np.column_stack([device, copy, night])

def fit_nb(X, y, alpha=1.0):
    prior = ...                # (a)  $P(y = 1)$ 
    th1 = ...                   # (b) Laplace-smoothed  $P(x_j = 1 \mid y = 1)$ 
    th0 = ...                   # and  $P(x_j = 1 \mid y = 0)$ 
    w = np.log(th1 / th0) - np.log((1 - th1) / (1 - th0))
    w0 = np.log(prior / (1 - prior)) + np.log((1 - th1) / (1 - th0)).sum()
    return w0, w

def fit_lr(X, y, steps=5000, lr=0.5):  # gradient ascent, as in Lecture 03
    Xb = np.column_stack([np.ones(len(X)), X]); w = np.zeros(Xb.shape[1])
    for i in range(steps):
```

1. Print w_0 , w and $P(\text{fraud} \mid x = (1, 1, 1))$ for both models.
2. Compare the weights of device and copy. Which model counts the device twice, and which shares the credit?
3. Which probability is closer to the fraction of frauds among the training rows with $x = (1, 1, 1)$?